Project: Worked examples
Title: Section properties
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Area, centroid, second moments of area (Ixx, Iyy, Ixy), principal axes and values, elastic and plastic section moduli, radii of gyration, torsion constant and (I-sections) warping constant for a parametric shape, an arbitrary polygon with holes, or a composite of placed parts. The x axis is horizontal (major axis of an I-section), y vertical.

WHAT IT CALCULATES	RELATION
Area and second moments of each outline, by Green's theorem	$A = \tfrac{1}{2}\oint(x\,dy - y\,dx)$; $I_{xx} = \int y^2\,dA$; $I_{yy} = \int x^2\,dA$; $I_{xy} = \int xy\,dA$
Second moments moved to the centroid by the parallel-axis theorem	$I_c = I - A\,d^2$
Principal axes	$\tan 2\theta = -2I_{xy}/(I_{xx} - I_{yy})$
Elastic section modulus	$W_{el} = I_{xx}/y_{max}$
Plastic section modulus, about the equal-area axis found by bisection	$W_{pl} = \lvert Q_{above}\rvert + \lvert Q_{below}\rvert$
Torsion constant by shape, a Saint-Venant series for a solid rectangle, $\pi d^4/32$ for a circle, Bredt for a closed thin-walled section, $\Sigma bt^3/3$ (plus a fillet term for rolled I) for an open section.	
Warping constant of an I-section	$I_w = I_{yy}\,(h - t_f)^2/4$

Basis: Properties of sections; torsion of solid and thin-walled sections[1](Appendix A, Table A.1; Ch. 10, Table 10.1); Torsional properties of rolled I-shapes[2](Appendix); Hot rolled steel channels, I and H sections: dimensions and masses[3]; Hot finished structural hollow sections: tolerances, dimensions and sectional properties[4].

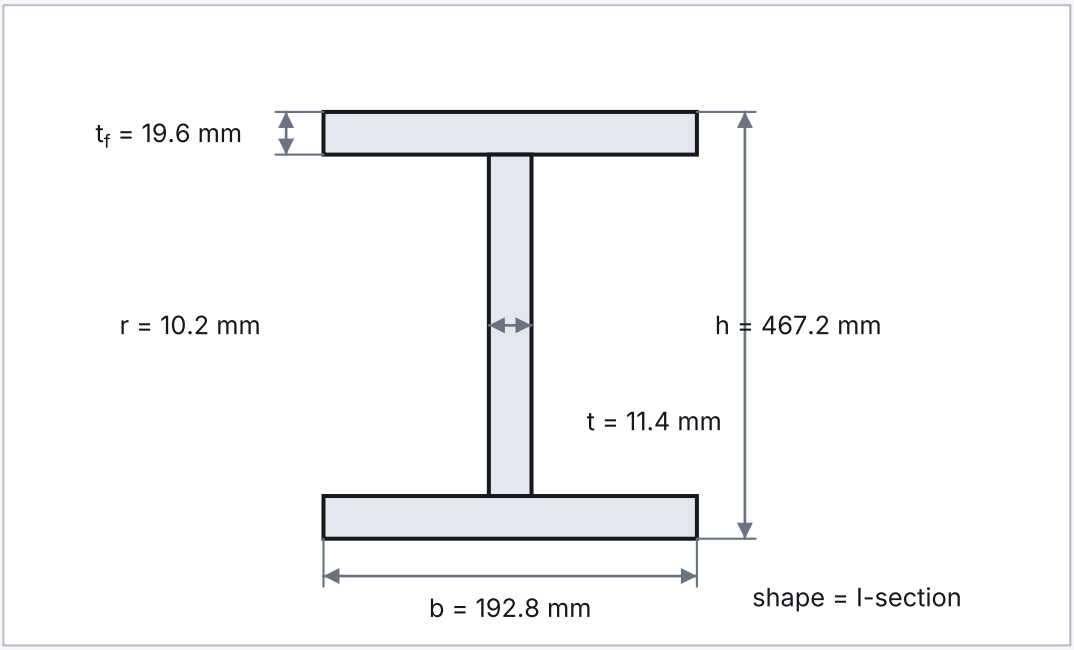


Figure 1 — A doubly symmetric I-section b by h, web t, flanges t_f, root radius r. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
shape	Shape	I-section		spec: BS EN 10365 UB 457×191×98	any
h	Depth h (or vertical leg)	467.2	mm	spec: BS EN 10365 UB 457×191×98	1 … 5000 mm
b	Width b (or horizontal leg)	192.8	mm	spec: BS EN 10365 UB 457×191×98	1 … 5000 mm
t	Wall / web / leg thickness t (t_w)	11.4	mm	spec: BS EN 10365 UB 457×191×98	0.1 … 500 mm
t_f	Flange thickness t_f	19.6	mm	spec: BS EN 10365 UB 457×191×98	0.1 … 500 mm
r	Root radius r (I, channel, tee, angle) or outer corner radius (hollow rectangle; blank = 1.5 t)	10.2	mm	spec: BS EN 10365 UB 457×191×98	0 … 500 mm

3. Assumptions

A1. A polygon outline is simple (no self-intersection); holes lie inside the outline and do not overlap. (Green's theorem integrates the signed loops as given.)
A2. Rolled-section root fillets are quarter circles of radius r; flange tips are square; hollow-section corners are quarter circles of outer radius r_o (default 1.5 t) and inner radius r_i (default 1.0 t). (Published section tables use the same idealisation.)
A3. Thin-walled torsion constants neglect the junction and tip effects except the fillet term kept for rolled I-sections. (Open sections: $\Sigma bt^3/3$; closed: Bredt.)

4. Working

[7] $A = \tfrac{1}{2} \oint (x\,dy - y\,dx)$ [1](Table A.1)

= Green's theorem over 108 vertices in 1 loop(s)

= 125.27 cm²

Area

[8] $x_c = \int x\,dA / A$

= Green's theorem over 108 vertices in 1 loop(s)

= 0 mm

Centroid x (from the shape origin)

[9] $y_c = \int y\,dA / A$

= Green's theorem over 108 vertices in 1 loop(s)

= 0 mm

Centroid y (from the shape origin)

[10] $I_{xx} = \int y^2\,dA - A\,y_c^2$

= Green's theorem over 108 vertices in 1 loop(s)

= 45728 cm⁴

Second moment of area about the centroidal x axis

[11] $I_{yy} = \int x^2\,dA - A\,x_c^2$

= Green's theorem over 108 vertices in 1 loop(s)

= 2347 cm⁴

Second moment of area about the centroidal y axis

[12] $I_{xy} = \int xy\,dA - A\,x_c\,y_c$

= Green's theorem over 108 vertices in 1 loop(s)

= 0 cm⁴

Product of inertia about the centroid

[13] $I_1 = (I_{xx} + I_{yy})/2 + \mathrm{sqrt}(((I_{xx} - I_{yy})/2)^2 + I_{xy}^2)$

= (45728 cm⁴ + 2347 cm⁴) / 2 + sqrt(((45728 cm⁴ - 2347 cm⁴) / 2)^2 + (0 cm⁴)^2)

= 45728 cm⁴

Major principal second moment

[14] $I_2 = (I_{xx} + I_{yy})/2 - \mathrm{sqrt}(((I_{xx} - I_{yy})/2)^2 + I_{xy}^2)$

= (45728 cm⁴ + 2347 cm⁴) / 2 - sqrt(((45728 cm⁴ - 2347 cm⁴) / 2)^2 + (0 cm⁴)^2)

= 2347 cm⁴

Minor principal second moment

[15] $\theta = \tfrac{1}{2}\,\mathrm{atan2}(-2\,I_{xy},\, I_{xx} - I_{yy})$

= ½ atan2(0, 43381)

= 0 deg

Angle from the x axis to the major principal axis (anticlockwise)

[16] $y_{top} = y_{max} - y_c$

= Green's theorem over 108 vertices in 1 loop(s)

= 233.6 mm

Extreme fibre above the centroid

[17] $y_{bot} = y_c - y_{min}$

= Green's theorem over 108 vertices in 1 loop(s)

= 233.6 mm

Extreme fibre below the centroid

[18] $x_{ext} = \max(x_{max} - x_c,\, x_c - x_{min})$

= Green's theorem over 108 vertices in 1 loop(s)

= 96.4 mm

Extreme fibre from the centroid, x

[19] $W_{el,x} = I_{xx} / \max(y_{top},\, y_{bot})$

= 45728 cm⁴ / max(233.6 mm, 233.6 mm)

= 1957.5 cm³

Elastic section modulus about x (governing fibre)

[20] $W_{el,y} = I_{yy} / x_{ext}$
$= 2347\ cm^4 / 96.4\ mm$
$= 243.47\ cm^3$
Elastic section modulus about y
[21] $W_{pl,x} = |Q_{above}| + |Q_{below}|$ about the equal-area axis
= equal-area axis at y = 0 mm
$= 2232.5\ cm^3$
Plastic section modulus about x
[22] $W_{pl,y} = |Q_{right}| + |Q_{left}|$ about the equal-area axis
= equal-area axis at x = 0 mm
$= 378.9\ cm^3$
Plastic section modulus about y
[23] $i_x = \sqrt{I_{xx} / A}$
$= \sqrt{45728\ cm^4 / 125.27\ cm^2}$
$= 19.106\ cm$
Radius of gyration about x
[24] $i_y = \sqrt{I_{yy} / A}$
$= \sqrt{2347\ cm^4 / 125.27\ cm^2}$
$= 4.3286\ cm$
Radius of gyration about y
[25] $\alpha_{pl,x} = W_{pl,x} / W_{el,x}$
$= 2232.5\ cm^3 / 1957.5\ cm^3$
$= 1.1404$
Shape factor about x
[26] $J = (2\ b\ t_f^3 + (h - 2\ t_f)\ t_w^3)/3 + 2\ \alpha\ D^4 - 0.42\ t_f^4$
$= b = 192.8,\ t_f = 19.6,\ h = 467.2,\ t_w = 11.4,\ r = 10.2\ mm$
$= 121.29\ cm^4$
Torsion constant
[27] $I_w = I_{yy}\ (h - t_f)^2 / 4$ ²(DG9)
$= 2347\ cm^4 \times (467.2 - 19.6)^2\ mm^2 / 4$
$= 1.1755\ dm^6$
Warping constant

5. Results

SYMBOL	VALUE	DESCRIPTION
A	125.27 cm²	Area
x_c	0 mm	Centroid x
y_c	0 mm	Centroid y
I_{xx}	45728 cm⁴	I_{xx}
I_{yy}	2347 cm⁴	I_{yy}
I_{xy}	0 cm⁴	I_{xy}
I_1	45728 cm⁴	I_1 (major principal)
I_2	2347 cm⁴	I_2 (minor principal)
θ	0 deg	Principal axis angle
$W_{el,x}$	1957.5 cm³	$W_{el,x}$
$W_{el,y}$	243.47 cm³	$W_{el,y}$
$W_{pl,x}$	2232.5 cm³	$W_{pl,x}$

SYMBOL	VALUE	DESCRIPTION
$W_{pl,y}$	378.9 cm³	$W_{pl,y}$
i_x	19.106 cm	i_x
i_y	4.3286 cm	i_y
$\alpha_{pl,x}$	1.1404	Shape factor, x
J	121.29 cm⁴	Torsion constant J
I_w	1.1755 dm⁶	Warping constant I_w

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

WARNING — thin-walled-torsion: J: open thin-walled sum with the rolled-section fillet term.

8. Verification

Template struct.section-properties@1.0.1 Content hash b7d59144d05267d4 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Seaburg PA, Carter CJ. Torsional Analysis of Structural Steel Members. Steel Design Guide 9. American Institute of Steel Construction; 1997.
3. British Standards Institution. Hot Rolled Steel Channels, I and H Sections—Dimensions and Masses. BS EN 10365. BSI.
4. British Standards Institution. Hot Finished Steel Structural Hollow Sections—Part 2: Tolerances, Dimensions and Sectional Properties. BS EN 10210-2. BSI.